

go into a theoretical description of the process, but whatever it be it seems certain that during recovery from tubercular disease many tubercles disappear entirely, and that they do not all become encapsuled or calcified, as seems to be the belief of many writers.

Perhaps the most frequent change which occurs is caseation of the tubercle, and where the tubercles are very numerous, or where tubercular infiltration is present, the caseation may be rapid, and we have the process termed "chronic suppuration." The process of caseation, especially when it is extensive, as in joints, is generally preceded by coalescence of tubercles and formation of the condition of tubercular infiltration. It frequently commences in the protoplasm of the epithelioid cells, but perhaps most often in the intercellular substance and in the processes of the giant cells. The protoplasm loses its homogeneous or finely granular appearance, and becomes coarsely granular, while the nuclei break up into fragments, which become distributed throughout the caseous material. [Specimens were shown illustrating various points in connection with caseous degeneration of tubercles, and illustrating the fact that the body of the giant cell is often one of the last structures to disappear.]

Lastly, in some cases the tubercle neither atrophies nor caseates rapidly, as above described, but the degenerative changes occur more slowly, and the degenerated material becomes infiltrated with calcareous salts, and can still be recognised after many years. [This was shown in a section of the lower end of the humerus from a case of disease of the elbow-joint which had been quiescent for fully ten years.]

This is the life-history of the individual tubercle, but another very important point in its life-history is that it is an infective nodule, that is to say that it does not remain single, but that once a tubercle develops, fresh ones spring up around the primary one. Indeed, it is in this way that the tubercular process spreads, the new tubercles coalescing with the old, and fresh ones appearing in the tissue around. And not only is the tubercle locally infective, it is also generally infective, a tubercular deposit in one part of the body being very apt to lead to tubercular deposit elsewhere. The tubercular process is also infective from man to man, from man to the lower animals, and from the lower animals to man. Mr. Barker has already mentioned a considerable number of the published cases of infection of wounds in man with tubercular material, and I need not, therefore, go into this matter. Nor need I go into the question of the infective agent in tubercular material; that has been conclusively shown by Koch and others to be the tubercle bacillus.

A further and very important point in the history of tubercles is that they are very irritating structures, and that the tissue in which they lie becomes the seat of simple inflammatory changes, which extend far beyond the limits of the tubercular disease, and produce very marked and destructive effects. We shall see ample evidence of this in the following specimens, and I need not therefore enlarge further on this matter at present.

To sum up, we see that tuberculosis is an infective disease due to the growth in the tissues of a parasitic micro-organism, the tubercle bacillus. It is characterised by the production of a special form of tissue, which may either present the form of nodules or of a more diffuse infiltration, the characteristic cellular elements of this tissue being the epithelioid cells. This tissue shows a great tendency to undergo a peculiar form of degeneration termed caseation, and it also excites and keeps up a condition of inflammation in the tissues around. The primary nodule does not remain single, but leads to fresh development of nodules both in the immediate vicinity and also frequently in distant parts. It is only by taking into consideration the histological characters of the tissue, the tendency to caseation, the presence of bacilli, the multiplicity of the nodules, etc., that we can come to a certain conclusion as to the tubercular nature of a disease. Of the various points which I have mentioned, those which I consider most important are the presence of epithelioid cells, the multiplicity of the nodules or the extent of the epithelioid infiltration, and the subsequent caseation. Where we have numerous and evidently spreading tubercles or large tracts of tubercular infiltration, and where caseation is occurring, I know of nothing else which this can indicate but tubercular disease. Limited collections of epithelioid and giant cells may occur around parasites of various kinds, but there is no tendency to extension of the process or to the formation of fresh epithelioid collections beyond the area of the parasite, while the constant tendency to multiplication is a characteristic of the tubercular process. I attach no importance to the demonstration

of tubercle bacilli as a means of diagnosing these bone and joint diseases. It would take up too much time to give what I believe to be the explanation of the apparently small numbers of tubercle bacilli in these joint affections; but I consider the above-mentioned histological characteristics sufficient to enable the diagnosis to be made without the aid of the tubercle bacillus.

We have thus arrived at a definite conception as to the histological structure of tubercle, and we may conclude that when we find tissue with the characteristics described above we have to do with tubercular disease. Now we find that the diseases in which this tissue is present are those which up till recently have been clinically known as strumous diseases; and, therefore, in speaking at the present time of surgical tuberculosis, we mean those diseases which clinically would be diagnosed as strumous.

There are many reasons why we must regard strumous diseases of bones and joints as tubercular, and, being tubercular, as due essentially to the development of the tubercle bacillus, but I cannot take up time in discussing them. I may, however, shortly summarise the facts as follows: In the first place, I have just referred to cases in which wounds in man have become infected with the tubercular virus, with the result that the various affections formerly called strumous, and among them strumous diseases of bones and joints, have developed. Further, in these strumous diseases the affected tissues show the presence of tubercles and tubercular tissue; tubercle bacilli are constantly present; these affections are intimately related to phthisis and other tubercular affections occurring elsewhere; the material derived from the diseased joints sets up tuberculosis in the lower animals, in the same manner as material from an undoubted tubercular source; and similar diseases can be induced in the lower animals by the introduction into bones and joints of tubercular material, and notably, as I have made out, of pure cultivations of tubercle bacilli. In describing the morbid anatomy of tubercular diseases of bones and joints I therefore describe the morbid anatomy of those affections known up till recently as "strumous diseases."

(To be continued.)

A DISCUSSION ON VACCINATION ERUPTIONS.

*In the Section of Dermatology at the Annual Meeting of the
British Medical Association, held in Birmingham,
July, 1890.*

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It is well that a question of such importance as vaccination eruptions should be discussed at the annual meeting of the British Medical Association, and it is particularly fitting that it should take place in this Section, as we have the good fortune to be presided over by one who has done so much to throw light into the dark corners of pathology. That we shall have the benefit of Mr. Hutchinson's great experience will do much to counteract the imperfections of the opening paper.

In order to remove any doubt that may exist as to my own opinion about vaccination, I may say that I am a firm believer in its value as the only real preventive of small-pox, and it is because I hold this view that I think every effort should be made to study the nature of the eruptions that are sometimes associated with vaccinia.

It is only by the knowledge gained from accurate diagnosis that we can ever hope to arrive at a position enabling us to speak authoritatively on the matter. So varied are the phenomena and so complex the questions involved that it is impossible to attempt to do justice to the whole subject in a short introductory paper. I would rather try to bring before you certain well-defined points for discussion, leaving out for the present many interesting collateral questions. The extreme paucity of the literature of this country may be ascribed, on the one hand, to a fear that any full account of unusual appearances may be detrimental to vaccination as a system, and, on the other hand, to the fact that the eruptions are fortunately rare. In answer, I would urge that it is one's duty to record, as far as possible, every deviation from what may be called the normal course.

Without further introduction I pass to the question of the